The documentation and process conversion measures necessary to comply with this revision shall be completed by 17 Feb 93.

INCH-POUND

MIL-S-19500/434B
17 November 1992
SUPERSEDING
MIL-S-19500/434A
1 September 1981

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, TRANSIENT VOLTAGE SUPPRESSOR
TYPES 1N5610 THROUGH 1N5613
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 1500 watt, peak, pulse power, silicon, transient, voltage suppressor diodes. Four levels of product assurance are provided for each device as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. Maximum ratings are as shown in columns 5 through 8 of table III herein, and as follows:

P_{PP} = 1500 W (see figure 3) at t_p = 1.0 ms.

$P_{M(AV)}$ = 3.0 W (derate at 20 mW/°C above T_A = +25°C) (see 6.3).

I_{FSM} = 150 A (pk) at t_p = 8.33 ms (T_A = +25°C).

-55°C ≤ T_{OP} ≤ +175°C (ambient), -55°C ≤ T_{STG} ≤ +175°C (ambient).

1.4 Primary electrical characteristics at T_A = +25°C. Primary electrical characteristics are shown in columns 2 and 4 of table III herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

V_C	Clamping voltage. The maximum peak voltage appearing across the device when subjected to the peak pulse current I_{PP}.
I_{PP}	Peak pulse current at a specified condition.
$I_{(BR)}$	Reverse breakdown current at a specified condition.
P_{PP}	Repetitive peak pulse power.
$\alpha V_{(BR)}$	Temperature coefficient of $V_{(BR)}$.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Metallurgical bond construction. Metallurgically bonded construction is required. The bonding metal shall have flow points above 260°C.

3.3.2 Lead finish. Lead finish shall be solderable as defined with MIL-STD-750, MIL-S-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Devices shall be marked as specified in MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the container.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	$T_{(high)}$ = +175°C	$T_{(high)}$ = +175°C
9	Not applicable	Not applicable
10	Not applicable	Not applicable
11	Not applicable	Not applicable
12	See 4.5.1	See 4.5.1
13	Interim electrical, delta, and group A, subgroup 2, electrical parameters not applicable for this screen (performed in screen 12). Subgroup 3 of table I herein.	Interim electrical, delta, and group A, subgroup 2, electrical parameters not applicable for this screen (performed in screen 12).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500. Subgroup 7: Condition for accelerated steady-state operation life are as follows: See 4.5.2, T_A = +100°C (min); 1 ms pulse only (see 4.5.3.b).

4.4.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV of MIL-S-19500. Subgroup 3: Condition for steady-state operation life are as follows: See 4.5.1, 1 ms pulse only (see 4.5.3.b).

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Condition for terminal strength are as follows: Lead tension: Test condition A; weight = 5 lbs; t = 15 ±3 s.

   Lead fatigue: Test condition E; weight = 8 oz.

b. Subgroup 6: Condition for steady-state operation life are as follows: See 4.5.1, 1 ms pulse only (see 4.5.3.b).

c. Subgroup 7: Condition for temperature coefficient of breakdown voltage are as follows: $I_{(BR)}$ = column 3 of table III, T_1 = +25°C ±3°C, T_2 = T_1 +100°C, LTPD = 10. All electrical measurements and test shall be performed twice, once in each direction.)

d. Subgroup 8: Condition for maximum peak pulse current are as follow: See 4.5.3a (20 μs pulse only) 10 pulses, LTPD 10.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Power burn-in (HTRB) and steady-state operation life test conditions. The test conditions and order of events shall be as follows:

a. Read and record I_D and $V_{(BR)}$ at T_A = +25°C. (This test is optional for JANTX and JANTXV devices during 100 percent screening).

b. Pulse in accordance with 4.5.3b, 20 times (screening and group B) and 100 times (group C) at T_A = +25°C.

c. Read I_D at T_A = +25°C, remove defective devices and record the number of failures. If step a is omitted during 100 percent screening, $V_{(BR)}$ shall be read and recorded at this step.

d. Apply the working peak reverse voltage (V_{WM}) (column 4 of table III) at T_A = +125°C as follows:

   1. 96 hours (JANTX and JANTXV) and 240 hours (JANS) for the screening test.

   2. 340 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

   3. 1,000 hours for group C steady-state operation life test.

e. Read and record I_D and $V_{(BR)}$ at T_A = +25°C. Devices with ΔI_D > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater, or $\Delta V_{(BR)}$ > ±2 percent (±5 percent for steady-state operation life) initial value shall be considered defective. Remove defective devices and record the number of failures. (For steady-state operation life, I_D maximum limit shall be two times the group A limit).

4.5.1.1 Group C steady-state operation life test (alternate procedure). When the group B 340 hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.5.1 with the following exceptions:

a. 4.5.1 step b shall be moved and performed following step e.

b. 4.5.1 step e shall be repeated after step b is performed.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reverse current	4016	DC method, $V_R = V_{WM}$ (column 4 of table III)	I_R		column 5	µA dc
Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table III	$V_{(BR)}$	column 2		V dc
Subgroup 3						
Minimum breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table III, $T_A = -55°C$	$V_{(BR)}$	column 9		V dc
Subgroup 4						
Clamping voltage maximum (pulsed) (see 4.5.4)		$t_p = 1.0$ ms (see 4.5.3.b), I_{PP} = column 7 of table III	V_C		column 6	V (pk)
Forward voltage	4011	$I_{FM} = 100$ A (pk), $t_p = 8.3$ ms duty (max), cycle = 4 pulses per minute (max)	V_{FM}		4.8	V (pk)
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge current	4066	$I_{FSM} = 150$ A (pk), one pulse, half sinewave (8.3 ms), $I_F = 0$, $V_{RWM} = 0$, $T_A = +25°C$				
End-point electrical measurements		See table II, steps 1 and 2				

1/ For sampling plan, see MIL-S-19500.

2/ Column references are to table III.

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TABLE II. Groups B and C electrical measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits 5/		Unit
		Method	Conditions		Min	Max	
1	Reverse current	4016	DC method, $V_R = V_{WM}$ column 4 of table III	I_R		column 5	µA dc
2	Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table III	$V_{(BR)}$	column 2		V dc
3	Forward voltage	4011	I_{FM} = 100 A (pk), t_p = 8.3 ms (max), duty cycle = 4 pulses per minute (max)	V_{FM}		4.8	V (pk)
4	Standby current	4016	DC method, $V_R = V_{WM}$ (column 4 of table III	ΔI_D 2/		100 percent of initial reading or 20 percent of column 5 of table III, whichever is greater	
5	Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table III	$\Delta V_{(BR)}$ 2/		±5 percent of initial value	
6	Clamping voltage		t_p = 1.0 ms (see 4.5.3.b), I_{PP} = column 7 of table III	V_C		column 6	V (pk)

1/ Devices which exceed the group A limits for this test shall not be accepted.

2/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, 3, and 6.

b. Subgroup 7, see table II herein, steps 1, 2, 3, 4, and 5.

3/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 6 see table II herein, steps 1, 2, 4, and 5.

4/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2 and 3, see table II herein, steps 1, 2, and 3 for JANS and steps 1 and 2 for JANTX and JANTXV.

b. Subgroup 8, see table II herein, steps 1 and 2 for all levels.

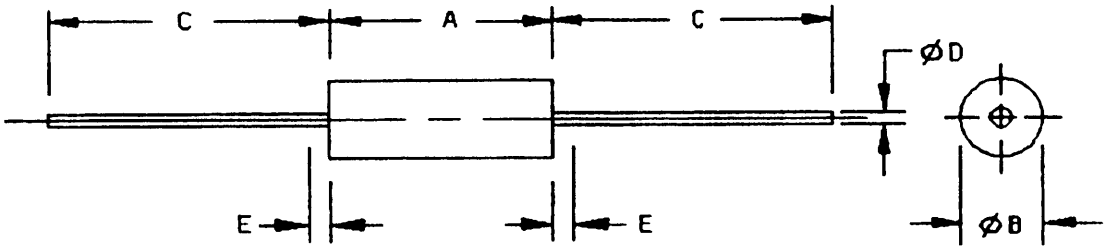
5/ Column references are to table III.

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TABLE III. Characteristics and Ratings.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7		Col 8	Col 9	Col 10
Type	Breakdown voltage $V_{(BR)}$ at $I_{(BR)}$ Min	Test current $I_{(BR)}$ $t_R \leq$ 300 ms cycle ≤ 2 percent	Working peak reverse voltage V_{WM}	Maximum reverse current I_R at V_{WM}	Maximum clamping voltage V_C at I_{PP} for t_p = 1 ms	Maximum peak pulse current (I_{PP})		Maximum temperature coefficient of $V_{(BR)}$ $\alpha V_{(BR)}$	Minimum breakdown voltage at $I_{(BR)}$ T_A = -55°C	Breakdown current $I_{(BR)}$ maximum dc current T_A = +25°C
						t_p = 20 µs t_r = 8 µs	t_p <= 1 ms t_r = 10 µs			
	V dc	mA dc	V (pk)	µA dc	V (pk)	A (pk)	A (pk)	α/°C	V dc	mA dc
1N5610	33.0	1	30.5	5	47.6	193	32.0	.093	30.2	75.0
1N5611	43.7	1	40.3	5	63.5	136	24.0	.094	40.0	53.0
1N5612	54.0	1	49.0	5	78.5	116	19.0	.096	48.5	43.0
1N5613	191.0	1	175.0	5	265.0	33	5.7	.100	172.0	12.5

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.225	.435	5.71	11.05	1,3
ØB	.150	.185	3.81	4.70	1
C	1.000		25.40		1,2
ØD	.027	.034	0.69	0.86	1,2
E		.050		1.27	1,4

NOTES:
1. Metric equivalents are given for general information only.
2. Both leads shall be within the specified limits.
3. See paragraph 3.4 for marking requirements.
4. The lead diameter is uncontrolled in this area.

FIGURE 1. Physical dimensions, type 1N5610 through 1N5613.

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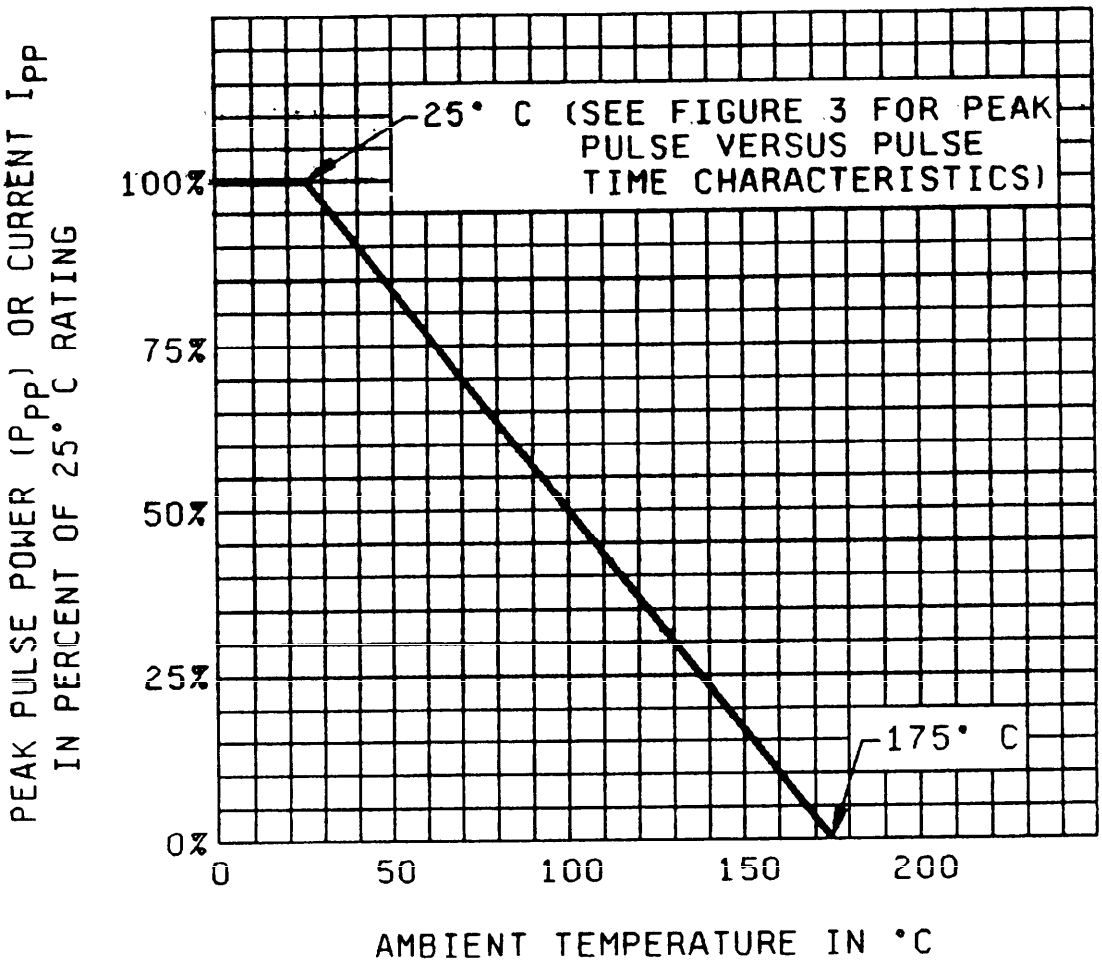


FIGURE 2. Derating curve.

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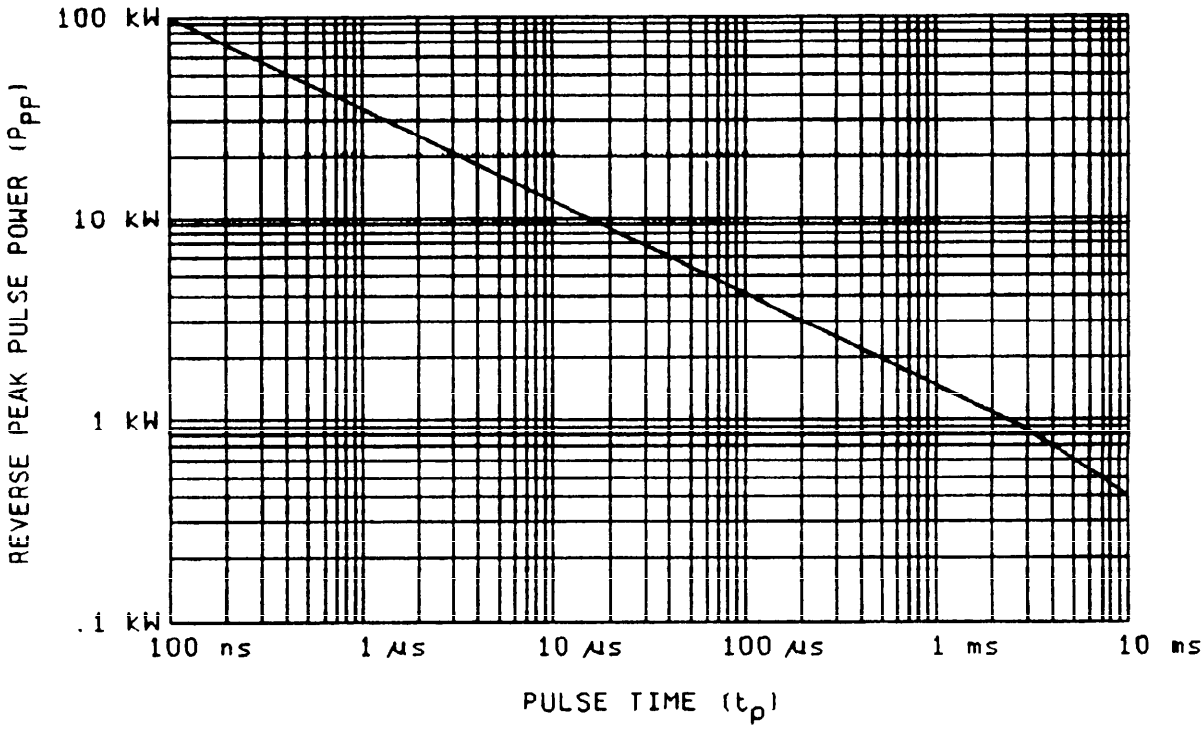
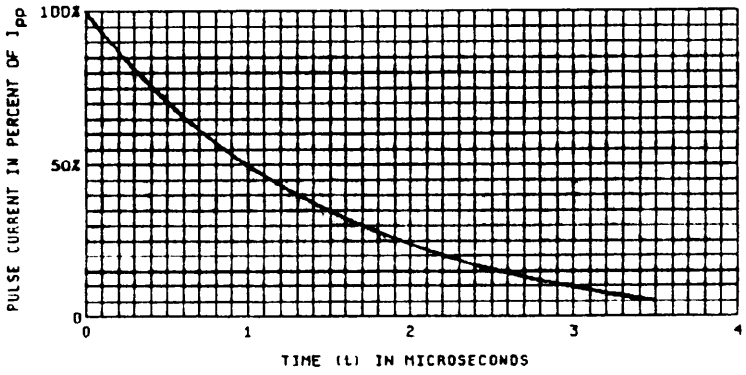


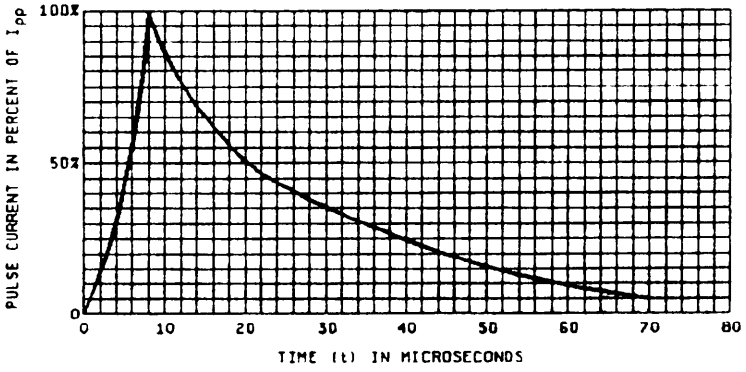
FIGURE 3. Peak pulse power vs. pulse time.

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NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of I_{PP}. (Rise time to 100 percent of I_{PP} = 10 μs).

FIGURE 4. Current impulse waveform.



NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of I_{PP}. (Rise time to 100 percent of I_{PP} = 8 μs).

FIGURE 5. Current impulse waveform.

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4.5.2 Accelerated steady-state operation life (JANS). This test shall be conducted with the devices subjected to the breakdown current specified in column 10 of table III for a total duration of 96 hours. At the beginning and end (with devices cooled to T_A = +25°C) of the test, the devices shall be subjected to pulse conditions at the rate of one pulse per minute (max) for 10 pulses in accordance with 4.5.3 as specified.

4.5.3 Maximum peak pulse current (I_{pp}). The peak pulse currents specified in column 7 of table III shall be applied in the reverse direction while simultaneously maintaining a reverse bias voltage of not less than the applicable voltage specified in column 4 of table III. The peak current shall be applied with a current VS time waveform as follows (1 pulse per minute maximum):

a. Pulse current shall reach 100 percent of I_{pp} at $t \leq 8\ \mu s$ and decay to 50 percent of I_{pp} at $t \geq 20\ \mu s$ for $t_p = 20\ \mu s$, (see figure 5).

b. Pulse current shall reach 100 percent of I_{pp} at $t \leq 10\ \mu s$ and decay to 50 percent of I_{pp} at $t \geq 1$ ms for $t_p = 1$ ms (see figure 4).

NOTE: Tolerance on time (t) shall be -0, +10 percent.

4.5.4 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in solicitation.

b. Lead finish as specified (see 3.3.2).

c. Type designation and product assurance level.

6.3 Steady-state power rating. This rating is not relevant for most applications.

6.4 Substitution information. For type numbers 1N5555 through 1N5558, formerly covered under this military specification are now covered under MIL-S-19500/500.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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